

DETAILED DIAGNOSTIC REPORT

Professional Building Diagnostics & Thermal Inspection Summary

Property Address:	123 Tech Blvd, Engineering City
Client Name:	ACME Corp
Inspection Date:	2026-07-07

1. Executive Summary

The subject property, Flat No. 103 and associated common areas, presents a Critical overall condition driven by systemic waterproofing failures and active water ingress. Our diagnostic analysis reveals a clear chain of causality: compromised tile joints and failed damp proof courses (DPC) in wet areas (Bathrooms/Kitchen) are allowing significant water migration through the structural slab, manifesting as active seepage in the Parking Area below and widespread rising dampness internally. The presence of efflorescence and biological growth confirms these are chronic, long-standing issues rather than isolated incidents. While the structural integrity of the RCC members currently appears stable with only moderate cracking, the persistent moisture environment poses an imminent risk of reinforcement corrosion and further degradation of interior finishes. Immediate intervention is required to halt water ingress, specifically targeting the Common Bathroom and external envelope, to prevent escalation into major structural repair liabilities.

2. Severity & Property Risk Assessment

Overall Property Severity: CRITICAL

Calculated Risk Score: 85 / 100

The property exhibits active water migration through the structural slab (OBS-006) affecting common areas, combined with widespread internal dampness (OBS-001, OBS-003) and biological growth (OBS-005). This indicates a systemic failure of the waterproofing envelope. While structural cracks are currently moderate, the sustained moisture exposure significantly accelerates the risk of rebar corrosion and concrete degradation, elevating the overall risk profile to Critical.

Building Health Indices:

Metric	Value
Overall Building Health Score	15%
Structural Health Index	65%
Waterproofing Health Index	20%
Interior Finish Health Index	40%
Moisture Risk Level	Critical
Estimated Total Cost Range	■181,000 - ■267,000

Area Action Priority Ranking:

- Priority 1: Parking Area (Active Seepage)
- Priority 2: Common Bathroom (Plumbing/Waterproofing Failure)
- Priority 3: External Wall (Envelope Breach)
- Priority 4: Bedrooms/Kitchen (Rising Damp/Efflorescence)
- Priority 5: RCC Members (Cracking)
- Priority 6: Master Bedroom Bathroom (Joint Failure)
- Priority 7: Flat 203 Bathrooms (Cosmetic)

3. Detailed Area-wise Observations

3.OBS-001 Hall - Rising Dampness

Significant dampness observed at the skirting level of the Hall in Flat No. 103. Visual evidence confirms moisture ingress at the wall-floor junction, consistent with capillary action or lateral water migration.

Severity:	Medium	Confidence:	Very High
Priority:	High	Timeline:	Within 7 Days
Root Cause:	Failure of the damp proof course (DPC) or rising groundwater due to compromised external plinth protection.	Estimated Cost:	■12,000 - ■18,000
Conflict Status:	No	Materials:	Chemical DPC Injection Fluid, Salt-Inhibiting Primer, Polymer Modified Cement, Waterproof Rendering Additive, Breathable Paint

Engineering Repair Procedure:

- 1. Remove existing skirting boards and damaged plaster up to 300mm above the visible damp line.
- 2. Allow the masonry to dry naturally or use forced air drying.
- 3. Inject a chemical damp proof course (DPC) at the base of the wall.
- 4. Apply a salt-inhibiting waterproof rendering slurry.
- 5. Re-plaster with polymer-modified cement mortar.
- 6. Re-install skirting and repaint with breathable emulsion.

AI Confidence Breakdown: Inspection Evidence: 100% | Thermal Evidence: 0% | Image Evidence: 100% | Overall: 95%

Evidence Sources:

- Inspection Report (Page 3)

3.OBS-002 Common Bathroom - Tile Hollowness and Plumbing Leakage

Observed gaps between tile joints, hollowness in floor/wall tiles, and indications of underlying plumbing issues in the Common Bathroom. Mild dampness is also present on the ceiling, suggesting vertical water migration from this unit or the unit above.

Severity:	High	Confidence:	High
Priority:	Critical	Timeline:	Within 24 Hours
Root Cause:	Degraded grout allowing water infiltration combined with potential leakage from concealed plumbing pipes or failed waterproofing membrane beneath tiles.	Estimated Cost:	■25,000 - ■35,000
Conflict Status:	No	Materials:	Flexible Waterproofing Membrane, Polymer Tile Adhesive, Epoxy Grout, PVC/CPVC Pipe Fittings (if repair needed), Tile Replacement Units

Engineering Repair Procedure:

- 1. Conduct a pressure test on concealed plumbing lines to isolate leaks.
- 2. Remove hollow and loose tiles completely.
- 3. Repair any identified plumbing leaks.
- 4. Clean the substrate and apply a two-component flexible waterproofing coating.
- 5. Re-lay tiles using high-grade polymer-modified tile adhesive.
- 6. Re-grout joints with epoxy or polyurethane grout for water tightness.

AI Confidence Breakdown: Inspection Evidence: 90% | Thermal Evidence: 0% | Image Evidence: 80% | Overall: 85%

Evidence Sources:

- Inspection Report (Page 3)
- Inspection Report (Page 4)

3.OBS-003 Bedroom, Master Bedroom, Kitchen - Widespread Skirting Dampness and Efflorescence

Consistent dampness observed at skirting levels across the Common Bedroom, Master Bedroom, and Kitchen. The Master Bedroom specifically exhibits efflorescence (salt deposits) on the wall surface, indicating long-term moisture evaporation and salt crystallization.

Severity:	High	Confidence:	Very High
Priority:	High	Timeline:	Within 7 Days
Root Cause:	Systemic failure of the building's damp proof course (DPC) or persistent lateral moisture ingress from the external environment affecting multiple rooms on the same floor level.	Estimated Cost:	■45,000 - ■60,000
Conflict Status:	No	Materials:	Crystalline Waterproofing Compound, Chemical DPC Injectant, Anti-Efflorescence Cleaner, Water-Resistant Cement, Alkali-Resistant Primer

Engineering Repair Procedure:

- 1. Strip plaster and paint from affected walls up to 1 meter height.
- 2. Treat masonry with anti-efflorescence acidic wash.
- 3. Install a new physical or chemical DPC barrier.
- 4. Apply crystalline waterproofing slurry to the bare masonry.
- 5. Re-plaster with water-resistant cement mortar mix.
- 6. Finish with alkali-resistant primer and breathable paint.

AI Confidence Breakdown: Inspection Evidence: 100% | Thermal Evidence: 0% | Image Evidence: 90% | Overall: 93%

Evidence Sources:

- Inspection Report (Page 3)
- Inspection Report (Page 4)
- Inspection Report (Page 4)

3.OBS-004 Master Bedroom Bathroom - Tile Joint Failure

Observed gaps between tile joints in the Master Bedroom Bathroom. While no hollowness was explicitly noted here, open joints pose a high risk of water seepage into the substrate.

Severity:	Medium	Confidence:	Medium
Priority:	Medium	Timeline:	Within 30 Days
Root Cause:	Shrinkage of cementitious grout over time or improper initial application, leading to loss of water tightness.	Estimated Cost:	■5,000 - ■8,000
Conflict Status:	No	Materials:	Epoxy Grout, Sanitary Silicone Sealant, Grout Removal Tool, Cleaning Solvent

Engineering Repair Procedure:

- 1. Rake out existing loose grout to a depth of at least 5mm.
- 2. Clean joints thoroughly with a vacuum and brush.
- 3. Apply epoxy or polyurethane-based grout.
- 4. Seal perimeter joints with sanitary silicone sealant.
- 5. Allow curing for 24 hours before water exposure.

AI Confidence Breakdown: Inspection Evidence: 80% | Thermal Evidence: 0% | Image Evidence: 0% | Overall: 60%

Evidence Sources:

- Inspection Report (Page 4)

3.OBS-005 External Wall - Structural Cracking, Duct Defect, and Biological Growth

Moderate cracks observed on the external wall near the Master Bedroom accompanied by a duct issue. Significant biological growth (algae, fungus, moss) is present on the surface, indicating prolonged moisture retention.

Severity:	High	Confidence:	Medium
Priority:	High	Timeline:	Within 7 Days
Root Cause:	Water ingress through cracked facade and compromised duct flashing, creating a damp environment conducive to biological colonization. Cracks may be due to thermal movement or settlement.	Estimated Cost:	■30,000 - ■45,000
Conflict Status:	No	Materials:	Fungicidal Wash, Elastomeric Crack Filler, Flashing Tape, Polyurethane Sealant, Exterior Waterproof Coating, Anti-Algal Paint

Engineering Repair Procedure:

- 1. Mechanically clean the facade to remove all biological growth using fungicidal wash.
- 2. V-out the cracks to create a key for repair material.
- 3. Fill cracks with elastomeric crack filler.
- 4. Repair/Seal the duct penetration point with waterproof flashing tape and sealant.
- 5. Apply a textured elastomeric waterproof coating over the entire affected external surface.
- 6. Repaint with anti-algal exterior emulsion.

AI Confidence Breakdown: Inspection Evidence: 90% | Thermal Evidence: 0% | Image Evidence: 0% | Overall: 70%

Evidence Sources:

- Inspection Report (Page 4)
- Inspection Report (Page 5)

3.OBS-006 Parking Area - Active Seepage from Slab

Observed leakage and seepage at the Parking Area ceiling located directly below Flat No. 103. This confirms water is migrating vertically through the structural slab from the units above.

Severity:	Critical	Confidence:	High
Priority:	Critical	Timeline:	Within 24 Hours
Root Cause:	Failure of waterproofing in the wet areas (Bathrooms/Kitchen) of Flat No. 103, allowing water to penetrate the RCC slab and emerge in the parking area below.	Estimated Cost:	■50,000 - ■75,000 (Includes source repair and ceiling restoration)
Conflict Status:	Yes	Materials:	Epoxy Injection Resin, Flexible Waterproofing Membrane, Cementitious Plug for active leaks, Restoration Plaster, Safety Signage/Barriers

Conflict Note: Visual inspection of the parking ceiling shows active seepage, yet thermal imaging data (OBS-010) shows a maximum temperature of 28.8°C and minimum of 20.1°C without explicitly highlighting a cold anomaly corresponding to the active leak. Active water leaks usually present as distinct cold spots due to evaporative cooling. The lack of a specific thermal anomaly call-out for this major leak suggests a potential gap in thermal analysis coverage for the parking ceiling or that the leak is intermittent.

Engineering Repair Procedure:

- 1. Immediate containment: Install drip collection trays in the parking area to prevent safety hazards.
- 2. Source identification: Perform flood testing in Flat 103 bathrooms/kitchen to pinpoint exact leak location.
- 3. Execute repair procedure defined in OBS-002 (Bathroom) or OBS-003 (Kitchen) depending on source.
- 4. Inject epoxy resin into the slab cracks from the parking side if structural cracks are the path.
- 5. Restore parking ceiling plaster and paint after leak is confirmed stopped.

AI Confidence Breakdown: Inspection Evidence: 95% | Thermal Evidence: 20% | Image Evidence: 0% | Overall: 75%

Evidence Sources:

- Inspection Report (Page 4)

3.OBS-007 Flat No. 203 Bathrooms - Tile Joint Gaps

Observed open tile joints in the Common and Master Bedroom Bathrooms of Flat No. 203. Note: This observation pertains to a different unit (203) but is recorded in the current inspection documentation.

Severity:	Low	Confidence:	Medium
Priority:	Low	Timeline:	Routine Maintenance
Root Cause:	Grout shrinkage or poor workmanship during tiling.	Estimated Cost:	■4,000 - ■6,000
Conflict Status:	No	Materials:	Water-Resistant Grout, Silicone Sealant, Grout Float

Engineering Repair Procedure:

- 1. Remove loose grout material.
- 2. Clean joints.
- 3. Apply new water-resistant grout.
- 4. Seal with silicone at wall-floor junctions.

AI Confidence Breakdown: Inspection Evidence: 80% | Thermal Evidence: 0% | Image Evidence: 0% | Overall: 60%

Evidence Sources:

- Inspection Report (Page 4)

3.OBS-008 RCC Members - Moderate Cracking

Moderate condition of cracks observed on RCC Column and Beam members. Specific locations not detailed, but implies structural elements.

Severity:	Medium	Confidence:	Medium
Priority:	High	Timeline:	Within 7 Days
Root Cause:	Thermal expansion/contraction, minor settlement, or plastic shrinkage during curing. Requires width measurement to rule out structural overload.	Estimated Cost:	■10,000 - ■20,000
Conflict Status:	No	Materials:	Crack Width Gauge, Low-Viscosity Epoxy Resin, Injection Ports, Elastomeric Bridge Coat

Engineering Repair Procedure:

- 1. Measure crack widths using a crack width gauge.
- 2. If widths < 0.3mm, apply surface sealing with elastomeric coating.
- 3. If widths > 0.3mm, perform epoxy injection to restore monolithic behavior.
- 4. Monitor cracks over 30 days for propagation.
- 5. Consult a structural engineer if cracks are active or widening.

AI Confidence Breakdown: Inspection Evidence: 85% | Thermal Evidence: 0% | Image Evidence: 0% | Overall: 65%

Evidence Sources:

- Inspection Report (Page 5)

3.OBS-009 General Interior - Thermal Anomaly

Thermal imaging recorded a maximum hotspot temperature of 28.8°C and a minimum coldspot of 20.1°C across inspected areas. The delta of 8.7°C suggests varying thermal mass or moisture content, though specific localization to defects requires correlation with visual data.

Severity:	Low	Confidence:	Very High
Priority:	Low	Timeline:	Routine Maintenance
Root Cause:	Temperature variations likely correspond to the damp areas identified in other observations (evaporative cooling causing cold spots) or solar gain on external walls (hot spots).	Estimated Cost:	■0
Conflict Status:	No	Materials:	

Engineering Repair Procedure:

- 1. No direct repair required for thermal readings.
- 2. Use thermal data to prioritize drying efforts in areas corresponding to cold spots (moisture).
- 3. Re-scan after repairs to verify elimination of thermal anomalies.

AI Confidence Breakdown: Inspection Evidence: 0% | Thermal Evidence: 100% | Image Evidence: 100% | Overall: 95%

Evidence Sources:

- Thermal Report (Page 1)

4. Final Engineering Assessment

The diagnostic investigation of 123 Tech Blvd reveals a property suffering from acute and systemic waterproofing failures. The primary driver of distress is water ingress, originating from failed tile joints and plumbing in wet areas (OBS-002, OBS-004) and exacerbated by a compromised external envelope (OBS-005). This water has penetrated the structural slab, causing active seepage in the parking area (OBS-006) and triggering widespread rising dampness and efflorescence in living spaces (OBS-001, OBS-003). While the observed structural cracking (OBS-008) is currently classified as moderate, the persistent saturation of concrete elements creates a high-risk environment for reinforcement corrosion, which could escalate structural concerns rapidly if left unaddressed. The conflict between visual seepage and lack of thermal anomalies suggests the leak may be intermittent or the thermal survey lacked sufficient resolution in the parking zone. A coordinated remediation strategy is mandatory: immediate source stopping in bathrooms, external sealing, and comprehensive internal drying/restoration. Delaying these repairs will result in exponential cost increases due to structural degradation and mold remediation requirements.

5. Recommended Corrective Actions

Immediate Actions

- **[Waterproofing & Safety]** Install safety barriers and drip trays in the Parking Area to mitigate slip/electrical hazards. Immediately isolate water supply to the Common Bathroom in Flat 103 and conduct pressure testing to identify active plumbing leaks. Begin emergency water extraction if standing water is present. (*Resolves: OBS-006, OBS-002*)
- **[External Envelope]** Apply temporary sealing to the external wall cracks and duct penetration points to prevent further rainwater ingress while permanent repairs are scheduled. Clean biological growth to halt spore propagation. (*Resolves: OBS-005*)

Short-term Actions

- **[Waterproofing & Restoration]** Execute full waterproofing remediation in the Common Bathroom (OBS-002) and Master Bedroom Bathroom (OBS-004). Strip and treat rising dampness in the Hall and Bedrooms (OBS-001, OBS-003) including chemical DPC injection and salt-inhibiting replastering. (*Resolves: OBS-002, OBS-004, OBS-001, OBS-003*)
- **[Structural Repair]** Measure and classify RCC cracks. Perform epoxy injection for structural cracks >0.3mm and elastomeric sealing for hairline cracks. Restore the Parking Area ceiling only after the source leak is confirmed fixed. (*Resolves: OBS-008, OBS-006*)

Long-term Preventive Actions

- **[Maintenance & Monitoring]** Apply a full elastomeric waterproof coating to the external facade. Establish a quarterly monitoring schedule for the repaired areas and Flat 203 to ensure joint integrity. Conduct annual thermal scans to detect early moisture recurrence. (*Resolves: OBS-005, OBS-007, OBS-009*)

6. Missing or Unclear Information

- **Measurement Data:** Specific crack width measurements (in mm) for RCC members (OBS-008) are missing, preventing definitive classification of structural vs. non-structural cracks.
- **Moisture Metrics:** Quantitative moisture meter readings (% relative humidity or % moisture content) for walls and slabs are not provided, relying solely on visual and thermal evidence.
- **Visual Documentation:** Photographic evidence is missing for several critical observations including the Parking Area seepage (OBS-006), External Wall cracks (OBS-005), and RCC cracks (OBS-008).
- **Structural Drawings:** Original structural drawings and reinforcement details are not available to assess the criticality of cracks in specific load-bearing columns or beams.
- **Building History:** Age of the property and history of previous repairs are unknown, which impacts the assessment of material degradation rates.